Gianluca Esposito

Contact Information

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Group Webpage
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Citizenship

Italy

Who am I

I am a Ph.D. student at Scuola Superiore Meridionale in Naples, and my main interest is Quantum Information theory, a very broad and versatile field which has many applications in quantum physics. My research activity involves topics like quantum complexity and quantum resource theory. One of my main interests is the understanding of the behavior of states under the action of quantum maps from a non-stabilizer resources point of view.

Research Interests

I am particularly interested in topics of quantum complexity. This has led me to the study of the resource theory of non-stabilizerness or "magic", which is a particular type of complexity essential for obtaining quantum advantage with respect to classical computation. However, the study of magic is involved in physical phenomena far beyond the field of quantum computation, and its interplay with other quantum resources like entanglement is highly non-trivial and yet to be fully understood. The behavior of magic in quantum evolutions (or more generally quantum maps) highly interests me, as it will provide tools to analyze both quantum many-body systems and quantum information-processing tasks.

Online identifiers

ORCID
ArXiv
Google Scholar

Current occupation

Ph.D. student in *"Mathematical and physical sciences for advanced materials and technologies"* at Scuola Superiore Meridionale, Naples

Member of the Quantum initiative of "Istituto Nazionale di Fisica Nucleare" (INFN), Naples Section.

Education

University of Naples Federico II, Naples, Italy

M.S., Physics, July 24th 2023

- Thesis Title: *Non-stabilizerness in Quantum Channels*
- Advisor: Professor Alioscia Hamma
- Final mark: 110/110 cum laude

B.S., Physics, November 21 2019

- Thesis Title: *Non existence of half-integer order Spherical Harmonics*
- Advisor: Professor Fedele Lizzi
- Final Mark: 110/110 cum laude

References Available to Contact

Alioscia Hamma, Ph.D. (e-mail: alioscia.hamma@unina.it; Phone: : +39 081-676495)

- Professor, Department of Physics
  University of Naples "Federico II"

◇ 2N34, Complesso Universitario di Monte Sant'Angelo Via Cintia - 80126 - Naples
⋆ *Dr. Hamma was my M.S. thesis advisor and he is my Ph.D. supervisor.*

Fedele Lizzi, Ph.D.
- Professor, Department of Physics
  University of Naples "Federico II"

◇ 2N27, Complesso Universitario di Monte Sant'Angelo Via Cintia - 80126 - Naples
⋆ *Dr. Lizzi was my B.S. thesis advisor.*

Awards

- 2023, University of Naples "Federico II", Physics Ph.D. program scholarship winner.
- 2023, University of Naples "Federico II", Quantum Technologies Ph.D. program scholarship winner.
- 2023, Scuola Superiore Meridionale in Naples, "Mathematical and physical sciences for advanced materials and technologies" (MPHS) Ph.D. program scholarship winner.
- 2025, "Iuvenum Quanta" at Stupor Quanti event call for Ph.D. students talks winner.

Academic Appointments

- **Visiting Student** February 2026 – September 2026
  Max Planck Institute for Quantum Optics – Theory Division, Garching, Germany

Publications **My top publications are marked with an asterisk (*)**

1. **Gianluca Esposito**, Georgios Styliaris. *Nonstabilizerness of quantum tensor network states is intractable in two dimensions*, arXiv preprint, 28/09/2026
   https://doi.org/10.48550/arXiv.2609.31459

2. **Gianluca Esposito**, Michele Viscardi, Alioscia Hamma. *Stabilizer entropy is trustworthy for mixed states*, arXiv preprint, 01/07/2026
   https://doi.org/10.48550/arXiv.2606.29443

3. Stefano Cusumano, **Gianluca Esposito**, Alioscia Hamma. *Probes of chaos over the Clifford group and approach to Haar values*, arXiv preprint, 01/04/2026
   https://doi.org/10.48550/arXiv.2603.29695

4. Halima Giovanna Ahmad, **Gianluca Esposito**, Viviana Stasino, Jovan Odavić, Carlo Cosenza, Alessandro Sarno, Pasquale Mastrovito, Michele Viscardi, Stefano Cusumano, Francesco Tafuri, Alioscia Hamma
   Experimental demonstration of non-local magic in a superconducting quantum processor
   arXiv preprint, 19/11/2025
   https://doi.org/10.48550/arXiv.2511.15576

5. **(*)** Daniele Iannotti, **Gianluca Esposito**, Lorenzo Campos Venuti, Alioscia Hamma
   Entanglement and Stabilizer entropies of random bipartite pure quantum states, Quantum 9, 1797, 21/07/2025
   https://doi.org/10.22331/q-2025-07-21-1797

6. Stefano Cusumano, Lorenzo Campos Venuti, Simone Cepollaro, **Gianluca Esposito**, Daniele Iannotti, Barbara Jasser, Jovan Odavić, Michele Viscardi, and Alioscia Hamma
   Non-stabilizerness and violations of CHSH inequalities, arXiv preprint, 04/04/2025
   https://doi.org/10.48550/arXiv.2504.03351

7. **Gianluca Esposito**, Simone Cepollaro, Luigi Cappiello, Alioscia Hamma
   Magic of discrete lattice gauge theories, International Journal of Geometric Methods in Modern Physics, 11/10/2024
   https://www.worldscientific.com/doi/10.1142/S0219887825500033

8. Simone Cepollaro, Goffredo Chirco, Gianluca Cuffaro, **Gianluca Esposito**, Alioscia Hamma
   Stabilizer entropy of quantum tetrahedra, Physical Review D, Vol.109, Issue 12 - 11/6/2024, 126008
   https://link.aps.org/doi/10.1103/PhysRevD.109.126008

9. **(*)** Lorenzo Leone, Salvatore F. E. Oliviero, **Gianluca Esposito**, Alioscia Hamma
   Phase transition in Stabilizer Entropy and efficient purity estimation, Physical Review A, Vol. 109, Issue 3 - 1/3/2024
   https://link.aps.org/doi/10.1103/PhysRevA.109.032403

Talks and posters

1. 17/03/2026, Max Planck Institute for Quantum Optics, Garching, Germany - Talk at Quantum Information Theory Division seminar: "Magic, Entanglement and Stabilizer Entropy"

2. 10/06/2025, Aula "Lorenzo Mangoni", Università degli Studi di Napoli "Federico II"- Talk at 2025 "Stupor Quanti" as "Iuvenum Quanta" speaker: "Cleansing algorithm for magic: an experiment with exponential speedup"

3. 21/03/2025, Dipartimento di Fisica "Ettore Pancini", Univeristà degli Studi di Napoli "Federico II" - Talk at Quantum Sciences and Technologies seminars: "Entanglement and stabilizer entropies in random pure states"

4. 18/12/2024, Dipartimento di Fisica "Ettore Pancini", Università degli studi di Napoli "Federico II" - Talk at "Recent Developments in Theoretical Physics": "Stabilizer Entropy as a proxy for magic".

5. 18/09/2024, Pizzo Calabro (VV) - Talk at 2024 "Italian Quantum Information Science" conference: "Quantum Resources with constraints".

Teaching Experience

- **December 2019-June 2022** Math and Physics tutor for high school and bachelor students.

Referee service

- *European Physics Journal Plus*
- *Quantum*
- *Chaos, solitons and Fractals*
- *Quantum Information Processing Conference 2027*

Outreach

- Volunteer at "Parla Potabile" outreach event organized in May 2022, in Naples, by PONYS, a young outreach association founded by "Federico II" physics students. The main goal of the event was to bring interest to physics and science topics to the general public, with special attention to children and teenagers.
- Volunteer at "Futuro Remoto" outreach event, also organized by PONYS, November 2019, Naples.

Programming Skills

Wolfram Mathematica, C, Python, LaTeX.